Data Dictionary

Rice (Cammeo and Osmancik) Classification – Raw and Engineered Features

1. Dataset Overview

Item	Detail
Dataset	Rice Dataset (Cammeo and Osmancik)
Source	Cinar, I. and Koklu, M. (2019). Classification of Rice Varieties Using Artificial Intelligence Methods. International Journal of Intelligent Systems and Applications in Engineering, 7(3), 188–194. https://doi.org/10.18201/ijisae.2019355381
Data origin	Images of rice grains processed with a computer vision system; 7 morphological features inferred per grain
Size	3,810 grains (rows) × 8 columns (7 numeric features + 1 class label)
Data quality	No missing values; no duplicate rows
Class balance	Cammeo: 1,630 grains (42.8%); Osmancik: 2,180 grains (57.2%) – mildly imbalanced
File format	ARFF (Rice_Cammeo_Osmancik.arff), loaded with scipy.io.arff

2. Raw Features

Raw features are the seven measurements supplied in the ARFF file. All size-related measurements are in pixels; Eccentricity and Extent are unitless.

#	Feature	Type	Unit	Description	Observed range (min – mean – max)
1	Area	Integer	pixels	Number of pixels within the boundary of the rice grain (grain size).	7,551 – 12,668 – 18,913
2	Perimeter	Real	pixels	Circumference of the grain, calculated from the distances between pixels around its boundary.	359.1 – 454.2 – 548.4
3	Major_Axis_Length	Real	pixels	Length of the longest line that can be drawn across the grain (main axis distance).	145.3 – 188.8 – 239.0
4	Minor_Axis_Length	Real	pixels	Length of the shortest line that can be drawn across the grain (small axis distance).	59.5 – 86.3 – 107.5
5	Eccentricity	Real	unitless (0–1)	How elongated the ellipse with the same moments as the grain is. 0 = perfect circle; closer to 1 = more elongated.	0.777 – 0.887 – 0.948

#	Feature	Type	Unit	Description	Observed range (min – mean – max)
6	Convex_Area	Integer	pixels	Pixel count of the smallest convex shell enclosing the region formed by the grain.	7,723 – 12,952 – 19,099
7	Extent	Real	unitless (0–1)	Ratio of the region formed by the grain to the pixels of its bounding box.	0.497 – 0.662 – 0.861

Target variable

Feature	Type	Description	Encoding
Class	Nominal	Rice variety of the grain: Cammeo or Osmancik.	LabelEncoder: Cammeo = 0, Osmancik = 1

3. Engineered Features

Engineered features are derived from the raw features. All are ratios, so they are unitless and describe grain shape rather than size.

Feature	Formula	Raw features needed	What it captures	Interpretation	Observed range (min – mean – max)	Status
Aspect_Ratio	Major_Axis_Length / Minor_Axis_Length	Major_Axis_Length, Minor_Axis_Length	Elongation of the grain	1 = round; larger = longer, slimmer grain	1.589 – 2.191 – 3.142	Explored (Experiment 1), dropped
Circularity	$(4\pi \times \text{Area}) / \text{Perimeter}^2$	Area, Perimeter	How close the outline is to a circle	1 = perfect circle; lower = more elongated or irregular boundary	0.620 – 0.770 – 0.903	Explored (Experiment 1), dropped
Solidity	Area / Convex_Area	Area, Convex_Area	How filled-in the grain is relative to its convex hull	1 = fully convex; lower = dents or irregular edges	0.926 – 0.978 – 0.990	Explored (Experiment 1), dropped
Perimeter_Area_Ratio	Perimeter / Area	Perimeter, Area	Boundary length per unit of area	Higher = smaller or more elongated grain, since Area grows faster than Perimeter	0.0286 – 0.0362 – 0.0489	Used in final custom set
Area_Axis_Product_Ratio	Area / (Major_Axis_Length × Minor_Axis_Length)	Area, Major_Axis_Length, Minor_Axis_Length	How well the grain fills the rectangle spanned by its axes	A perfect ellipse gives $\pi/4 \approx 0.785$, so values near 0.785 indicate a near-elliptical shape	0.758 – 0.775 – 0.784	Used in final custom set

Rationale for feature engineering

Area, Perimeter, Convex_Area and Major_Axis_Length are highly correlated (for example, Area and Convex_Area have r ≈ 1.00). Ratio features were created to retain the shape information while reducing multicollinearity among predictors.

4. Feature Sets Used

Feature set	Features	Count
Original	Area, Perimeter, Major_Axis_Length, Minor_Axis_Length, Eccentricity, Convex_Area, Extent	7
Engineered – Experiment 1	Area, Extent, Aspect_Ratio, Circularity, Solidity (Perimeter, Major_Axis_Length, Minor_Axis_Length, Eccentricity and Convex_Area dropped)	5
Final custom set	Eccentricity, Extent, Perimeter_Area_Ratio, Area_Axis_Product_Ratio	4

5. Preprocessing Applied to the Final Custom Features

Feature	Transformation	Reason
Eccentricity	RobustScaler	Small share of IQR-flagged outliers
Extent	RobustScaler	Low skewness; weakly discriminative
Perimeter_Area_Ratio	RobustScaler	Small share of IQR-flagged outliers
Area_Axis_Product_Ratio	Box-Cox, then RobustScaler	Moderate negative skew; all values are positive, so Box-Cox is applicable

Note: Observed ranges are computed on the full dataset of 3,810 rows. Scalers and transformations are fitted on training rows only.